TALL SOM FORTELLER

GaussSuppression

SDC Tools

CRAN Task Views OfficialStatistics

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Co-funded by the European Union

Package ‘GaussSuppression’

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Statistisk sentralbyrå
Statistics Norway

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GaussSuppression: An R package for Tabular Data Suppression

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https://cran.r-project.org/package=GaussSuppression
https://github.com/statisticsnorway/GaussSuppression

STATISTICS NORWAY

USER GROUP SDC WORKSHOP IN PARIS

20TH SEPTEMBER 2022

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News on CRAN since the 2022 workshop

GaussSuppression 0.6.0

GaussSuppression 0.7.0

GaussSuppression 0.8.0

GaussSuppression 0.8.3

GaussSuppression 0.8.5

GaussSuppression 0.8.8

GaussSuppression 0.9.0

GaussSuppression 0.9.2

GaussSuppression 0.9.5

GaussSuppression 0.9.6

GaussSuppression 1.1.0

GaussSuppression 1.1.5

GaussSuppression 1.2.0

GaussSuppression 1.2.3

GaussSuppression 1.3.0

- Major update of `SuppressKDisclosure()` and related functions.
  - The description of `SuppressKDisclosure()` is now: *Frequency table suppression for targeted attribute disclosure protection.*
  - New parameter `identifying` in `SuppressKDisclosure()` to specify information that an intruder may already know.
  - New parameter `sensitive` in `SuppressKDisclosure()` to specify information considered unacceptable to disclose.
  - Further specification of identifying–sensitive relations may be provided via the `default_targeting()` parameters `targeting_include` and `targeting_exclude`.
  - The mechanism underlying these parameters is implemented through a new `targeting` parameter in `KDisclosurePrimary()` (see its documentation for details). The `targeting` specification may be generated by a function (default: `default_targeting()`), similar to other user-supplied functions. The parameters `identifying` and `sensitive` are included as explicit arguments in `SuppressKDisclosure()`, but are in fact arguments of `default_targeting()`.
  - The default `singletonMethod` for `SuppressKDisclosure()` has been updated (via `PackageSpecs()`) from

Contents

Defining tables and SuppressSmallCounts()

SuppressDominantCells()

Interval suppression

SuppressLinkedTables()

tables_by_formulas()

Functionality that corresponds to SSBtools::model_aggregate()

secondary suppression

Defining tables and SuppressSmallCounts()

secondary suppression

age	Denmark	France	USA	Total
young	1	2	4	7
old	1	6	3	10
Total	2	8	7	17

- One of many possible solutions
- The GaussSuppression solution

secondary suppression

age	Denmark	France	USA	Total
young	1	2	4	**7**
old	1	6	3	**10**
Total	2	8	7	**17**

- The optimal solution
- Often too computationally demanding in practice

The Gauss algorithm

age	Denmark	France	USA	Total
young	1	2	4?	7?
old	1	6?	3?	10?
Total	2	8?	7?	17?

- uses a priority order
  - The frequencies in descending order

The Gauss algorithm

age	Denmark	France	USA	Total
young	1	2	4?	7?
old	1	6?	3?	10?
Total	2	8?	7?	17

- uses a priority order
  - The frequencies in descending order

The Gauss algorithm

age	Denmark	France	USA	Total
young	1	2	4?	7?
old	1	6?	3?	10
Total	2	8?	7?	17

- uses a priority order
  - The frequencies in descending order

The Gauss algorithm

age	Denmark	France	USA	Total
young	1	2	4?	7?
old	1	6?	3?	10
Total	2	8	7?	17

- uses a priority order
  - The frequencies in descending order

The Gauss algorithm

age	Denmark	France	USA	Total
young	1	2	4?	7
old	1	6?	3?	10
Total	2	8	7?	17

- uses a priority order
  - The frequencies in descending order

The Gauss algorithm

age	Denmark	France	USA	Total
young	1	2	4?	7
old	1	6?	3?	10
Total	2	8	7	17

- uses a priority order
  - The frequencies in descending order

The Gauss algorithm

age	Denmark	France	USA	Total
young	1	2	4?	7
old	1	6	3?	10
Total	2	8	7	17

- uses a priority order
  - The frequencies in descending order

The Gauss algorithm

age	Denmark	France	USA	Total
young	1	2	4	7
old	1	6	3?	10
Total	2	8	7	17

- uses a priority order
  - The frequencies in descending order

The Gauss algorithm

age	Denmark	France	USA	Total
young	1	2	4	7
old	1	6	3	10
Total	2	8	7	17

- uses a priority order
  - The frequencies in descending order

Hierarchical table

	nation	continent	age	freq
1	Denmark	Europe	old	1
2	Denmark	Europe	young	1
3	France	Europe	old	6
4	France	Europe	young	2
5	USA	America	old	3
6	USA	America	young	4

age	Denmark	France	USA	America	Europe	Total
young	1	2	4	**4**	**3**	**7**
old	1	6	3	**3**	**7**	**10**
Total	2	8	**7**	**7**	**10**	**17**

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```r
SuppressSmallCounts(
 data = microdata,
 dimVar = c("nation","continent","age"),
 maxN = 2
)
```

age	Denmark	France	USA	America	Europe	Total
young	1	2	4	**4**	**3**	**7**
old	1	6	3	**3**	**7**	**10**
Total	2	8	**7**	**7**	**10**	**17**

```r
microdata <-
  SSBtoolsData("paris2025_micro")
```

```
      nation continent   age
 1   Denmark    Europe   old
 2   Denmark    Europe young
 3    France    Europe   old
 4    France    Europe   old
 5    France    Europe   old
 6    France    Europe   old
 7    France    Europe   old
 8    France    Europe   old
 9    France    Europe young
 10   France    Europe young
 11      USA   America   old
 12      USA   America   old
 13      USA   America   old
 14      USA   America young
 15      USA   America young
 16      USA   America young
 17      USA   America young
```

```
SuppressSmallCounts(
 data = microdata,
 dimVar = c("nation","continent","age"),
 maxN = 2
)
```

	nation	age	freq	primary	suppressed
1	Total	Total	17	FALSE	FALSE
2	Total	old	10	FALSE	FALSE
3	Total	young	7	FALSE	FALSE
4	America	Total	7	FALSE	FALSE
5	America	old	3	TRUE	TRUE
6	America	young	4	FALSE	TRUE
7	Europe	Total	10	FALSE	FALSE
8	Europe	old	7	FALSE	TRUE
9	Europe	young	3	TRUE	TRUE
10	Denmark	Total	2	TRUE	TRUE
11	Denmark	old	1	TRUE	TRUE
12	Denmark	young	1	TRUE	TRUE
13	France	Total	8	FALSE	TRUE
14	France	old	6	FALSE	FALSE
15	France	young	2	TRUE	TRUE
16	USA	Total	7	FALSE	FALSE
17	USA	old	3	TRUE	TRUE
18	USA	young	4	FALSE	TRUE

```
microdata <-
  SSBtoolsData("paris2025_micro")
```

	nation	continent	age
1	Denmark	Europe	old
	Denmark	Europe	young
	[illegible]	Europe	old
		Europe	old
		[illegible]urope	old
		[illegible]	old
		[illegible]	[illegible]ld
11			
12	[illegible]		
13	USA		
14	USA	Amer[illegible]	
15	USA	America [illegible]	
16	USA	America young	
17	USA	America young	

age	Denmark	France	USA	America	Europe	Total
young	1	2	4	4	3	7
old	1	6	3	3	7	10
Total	2	8	7	7	10	17

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```
SuppressSmallCounts(
 data = microdata,
 formula = ~ age*(nation+continent),
 maxN = 2
)
```

	nation	age	freq	primary	suppressed
1	Total	Total	17	FALSE	FALSE
2	Total	old	10	FALSE	FALSE
3	Total	young	7	FALSE	FALSE
4	America	Total	7	FALSE	FALSE
5	America	old	3	TRUE	TRUE
6	America	young	4	FALSE	TRUE
7	Europe	Total	10	FALSE	FALSE
8	Europe	old	7	FALSE	TRUE
9	Europe	young	3	TRUE	TRUE
10	Denmark	Total	2	TRUE	TRUE
11	Denmark	old	1	TRUE	TRUE
12	Denmark	young	1	TRUE	TRUE
13	France	Total	8	FALSE	TRUE
14	France	old	6	FALSE	FALSE
15	France	young	2	TRUE	TRUE
16	USA	Total	7	FALSE	FALSE
17	USA	old	3	TRUE	TRUE
18	USA	young	4	FALSE	TRUE

```
microdata <-
  SSBtoolsData("paris2025_micro")
```

	nation	continent	age
1	Denmark	Europe	old
	Denmark	Europe	young
	[illegible]	Europe	old
		Europe	old
		[illegible]urope	old
		[illegible]	old
		[illegible]	[illegible]ld
11			
12	[illegible]		
13	USA		
14	USA	Amer[illegible]	
15	USA	America [illegible]	
16	USA	America young	
17	USA	America young	

age	Denmark	France	USA	America	Europe	Total
young	1	2	4	4	3	7
old	1	6	3	3	7	10
Total	2	8	7	7	10	17

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```
SuppressSmallCounts(
 data = microdata,
 hierarchies = my_hierarchies
 maxN = 2
)
```

	nation	age	freq	primary	suppressed
1	Total	Total	17	FALSE	FALSE
2	Total	old	10	FALSE	FALSE
3	Total	young	7	FALSE	FALSE
4	America	Total	7	FALSE	FALSE
5	America	old	3	TRUE	TRUE
6	America	young	4	FALSE	TRUE
7	Europe	Total	10	FALSE	FALSE
8	Europe	old	7	FALSE	TRUE
9	Europe	young	3	TRUE	TRUE
10	Denmark	Total	2	TRUE	TRUE
11	Denmark	old	1	TRUE	TRUE
12	Denmark	young	1	TRUE	TRUE
13	France	Total	8	FALSE	TRUE
14	France	old	6	FALSE	FALSE
15	France	young	2	TRUE	TRUE
16	USA	Total	7	FALSE	FALSE
17	USA	old	3	TRUE	TRUE
18	USA	young	4	FALSE	TRUE

```
microdata <-
  SSBtoolsData("paris2025_micro")
```

	nation	continent	age
1	Denmark	Europe	old
	Denmark	Europe	young
	[illegible]	Europe	old
		Europe	old
		[illegible]urope	old
		[illegible]	old
		[illegible]	[illegible]ld
11			
12	[illegible]		
13	USA		
14	USA	Amer[illegible]	
15	USA	America [illegible]	
16	USA	America young	
17	USA	America young	

age	Denmark	France	USA	America	Europe	Total
young	1	2	4	4	3	7
old	1	6	3	3	7	10
Total	2	8	7	7	10	17

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my_hierarchies - alternatives

```
$nation
  levels   codes
1      @   Total
2     @@ America
3    @@@     USA
4     @@  Europe
5    @@@ Denmark
6    @@@  France

$age
  levels codes
1      @ Total
2     @@   old
3     @@ young
```

```
$nation
[1] "America"  "@USA"     "Europe"   "@Denmark" "@France"

$age
[1] "old"   "young"
```

```
$nation
     from      to
1 America   Total
2     USA America
3  Europe   Total
4 Denmark  Europe
5  France  Europe

$age
   from    to
1   old Total
2 young Total
```

```
$nation
      root    leaf level
    <char>  <char> <num>
1:   Total   Total     1
2:   Total America     2
3: America     USA     3
4:   Total  Europe     2
5:  Europe Denmark     3
6:  Europe  France     3

$age
     root   leaf level
   <char> <char> <num>
1:  Total  Total     1
2:  Total    old     2
3:  Total  young     2
```

Linked tables

age	Denmark	France	USA	America	Europe	Total
young				4	3	7
old				3	7	10
Total	2	8	7	7	10	17

```r
SuppressSmallCounts(
    data = microdata,
    formula = ~nation + age * continent,
    maxN = 3
)
```

age	Denmark	France	USA	America	Europe	Total
young				4	3	7
old				3	7	10
Total	2	8	7	7	10	17

```r
microdata <-
  SSBtoolsData("paris2025_micro")
```

```
      nation continent   age
 1   Denmark    Europe   old
 2   Denmark    Europe young
 3    France    Europe   old
 4    France    Europe   old
 5    France    Europe   old
 6    France    Europe   old
 7    France    Europe   old
 8    France    Europe   old
 9    France    Europe young
10    France    Europe young
11       USA   America   old
12       USA   America   old
13       USA   America   old
14       USA   America young
15       USA   America young
16       USA   America young
17       USA   America young
```

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```r
SuppressSmallCounts(
    data = microdata,
    formula = ~nation + age * continent,
    maxN = 3
)
```

	nation	age	freq	primary	suppressed
1	Total	Total	17	FALSE	FALSE
2	Denmark	Total	2	TRUE	TRUE
3	France	Total	8	FALSE	TRUE
4	USA	Total	7	FALSE	FALSE
5	Total	old	10	FALSE	FALSE
6	Total	young	7	FALSE	FALSE
7	America	Total	7	FALSE	FALSE
8	Europe	Total	10	FALSE	FALSE
9	America	old	3	TRUE	TRUE
10	Europe	old	7	FALSE	TRUE
11	America	young	4	FALSE	TRUE
12	Europe	young	3	TRUE	TRUE

```r
microdata <-
  SSBtoolsData("paris2025_micro")
```

	nation	continent	age
1	Denmark	Europe	old
	Denmark	Europe	young
	ance	Europe	old
		Europe	old
		Europe	old
			old
			old
11			
12	US		
13	USA		
14	USA	Amer	
15	USA	America	y
16	USA	America	young
17	USA	America	young

age	Denmark	France	USA	America	Europe	Total
young				4	3	7
old				3	7	10
Total	2	8	7	7	10	17

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formula - examples

```
~ nation + age * continent
```

```
~ age * (nation + continent)
```

The first group of hypercubes for the 2021 census.

```
~ sex * (age_l + age_m + age_h) * (lms_l + lms_h) +
  sex * (age_l + age_m + age_h) * (hst_l + hst_m + hst_h) +
  sex * (age_l + age_m + age_h) * (fst_l + fst_m + fst_h) +
  sex * (lms_l + lms_h) * (hst_l + hst_m + hst_h)
```

```
> microdata <-
    SSBtoolsData("paris2025_micro")
>
> microdata
    nation continent   age income
1  Denmark    Europe   old   1220
2  Denmark    Europe young    160
3   France    Europe   old   1450
4   France    Europe   old   1860
5   France    Europe   old   3030
6   France    Europe   old   5880
7   France    Europe   old   6570
8   France    Europe   old   9370
9   France    Europe young    460
10  France    Europe young   1300
11     USA   America   old    530
12     USA   America   old   3980
13     USA   America   old   8500
14     USA   America young    280
15     USA   America young    700
16     USA   America young   3640
17     USA   America young   4680
```

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```
> out_1 <- SuppressSmallCounts(
    data = microdata,
    hierarchies =
      list(nation = c("America", "@USA", "Europe", "@Denmark", "@France"),
           age = c("old", "young")),
    maxN = 2
  )
[preAggregate 17*4->6*3]
[extend0 6*3->6*3]
GaussSuppression_anySum: ..........
> out_1
    nation   age freq primary suppressed
1    Total Total   17   FALSE      FALSE
2    Total   old   10   FALSE      FALSE
3    Total young    7   FALSE      FALSE
4  America Total    7   FALSE      FALSE
5  America   old    3   FALSE      FALSE
6  America young    4   FALSE      FALSE
7   Europe Total   10   FALSE      FALSE
8   Europe   old    7   FALSE      FALSE
9   Europe young    3   FALSE      FALSE
10 Denmark Total    2    TRUE       TRUE
11 Denmark   old    1    TRUE       TRUE
12 Denmark young    1    TRUE       TRUE
13  France Total    8   FALSE       TRUE
14  France   old    6   FALSE       TRUE
15  France young    2    TRUE       TRUE
16     USA Total    7   FALSE      FALSE
17     USA   old    3   FALSE      FALSE
18     USA young    4   FALSE      FALSE
```

```
> df <- data.frame(
    nation = c("Denmark", "Denmark", "France", "France", "USA", "USA"),
    continent = c("Europe", "Europe", "Europe", "Europe", "America", "America"),
    age = c("young", "old", "young", "old", "young", "old"),
    freq = c(1, 1, 2, 6, 4, 3))
>
> df
   nation continent   age freq
1 Denmark    Europe young    1
2 Denmark    Europe   old    1
3  France    Europe young    2
4  France    Europe   old    6
5     USA   America young    4
6     USA   America   old    3
```

```
>
```

```
> out_2 <- SuppressSmallCounts(
    data = df,
    formula = ~nation + age * continent,
    maxN = 2,
    freqVar = "freq"
  )
[extend0 6*4->6*4]
GaussSuppression_anySum: ..........
> out_2
    nation   age freq primary suppressed
1    Total Total   17   FALSE      FALSE
2  Denmark Total    2    TRUE       TRUE
3   France Total    8   FALSE       TRUE
4      USA Total    7   FALSE      FALSE
5    Total   old   10   FALSE      FALSE
6    Total young    7   FALSE      FALSE
7  America Total    7   FALSE      FALSE
8   Europe Total   10   FALSE      FALSE
9  America   old    3   FALSE      FALSE
10  Europe   old    7   FALSE      FALSE
11 America young    4   FALSE      FALSE
12  Europe young    3   FALSE      FALSE
```

```
> df
   nation continent   age freq
1 Denmark    Europe young    1
2 Denmark    Europe   old    1
3  France    Europe young    2
4  France    Europe   old    6
5     USA   America young    4
6     USA   America   old    3
```

```
>
```

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```
> # With base R pipe operator (similar to %>% )
> out_3 <- microdata |> SuppressSmallCounts(
    formula = ~nation + age * continent,
    maxN = 2)
[preAggregate 17*4->6*4]
[extend0 6*4->6*4]
GaussSuppression_anySum: ..........
> out_3
    nation   age freq primary suppressed
1    Total Total   17   FALSE      FALSE
2  Denmark Total    2    TRUE       TRUE
3   France Total    8   FALSE       TRUE
4      USA Total    7   FALSE      FALSE
5    Total   old   10   FALSE      FALSE
6    Total young    7   FALSE      FALSE
7  America Total    7   FALSE      FALSE
8   Europe Total   10   FALSE      FALSE
9  America   old    3   FALSE      FALSE
10  Europe   old    7   FALSE      FALSE
11 America young    4   FALSE      FALSE
12  Europe young    3   FALSE      FALSE

>
```

pPercent

SuppressDominantCells()

```r
SuppressDominantCells(
  data = microdata,
  formula = ~nation + age * continent,
  dominanceVar =  "income",
  pPercent = 10
)
```

	nation	age	income	primary	suppressed
1	Total	Total	53610	FALSE	FALSE
2	Denmark	Total	1380	TRUE	TRUE
3	France	Total	29920	FALSE	TRUE
4	USA	Total	22310	FALSE	FALSE
5	Total	old	42390	FALSE	FALSE
6	Total	young	11220	FALSE	FALSE
7	America	Total	22310	FALSE	FALSE
8	Europe	Total	31300	FALSE	FALSE
9	America	old	13010	TRUE	TRUE
10	Europe	old	29380	FALSE	TRUE
11	America	young	9300	FALSE	TRUE
12	Europe	young	1920	FALSE	TRUE

```r
microdata <-
  SSBtoolsData("paris2025_micro")
```

	nation	continent	age	income
1	Denmark	Europe	old	1220
2	Denmark	Europe	young	160
3	France	Europe	old	1450
4	France	Europe	old	1860
5	France	Europe	old	3030
6	France	Europe	old	5880
7	France	Europe	old	6570
8	France	Europe	old	9370
9	France	Europe	young	460
10	France	Europe	young	1300
11	USA	America	old	530
12	USA	America	old	3980
13	USA	America	old	8500
14	USA	America	young	280
15	USA	America	young	700
16	USA	America	young	3640
17	USA	America	young	4680

```r
SuppressDominantCells(
  data = microdata,
  formula = ~nation + age * continent,
  dominanceVar =  "income",
  k = c(80, 90)
)
```

	nation	age	income	primary	suppressed
1	Total	Total	53610	FALSE	FALSE
2	Denmark	Total	1380	TRUE	TRUE
3	France	Total	29920	FALSE	TRUE
4	USA	Total	22310	FALSE	FALSE
5	Total	old	42390	FALSE	FALSE
6	Total	young	11220	FALSE	FALSE
7	America	Total	22310	FALSE	FALSE
8	Europe	Total	31300	FALSE	FALSE
9	America	old	13010	TRUE	TRUE
10	Europe	old	29380	FALSE	TRUE
11	America	young	9300	FALSE	TRUE
12	Europe	young	1920	TRUE	TRUE

```r
microdata <-
  SSBtoolsData("paris2025_micro")
```

	nation	continent	age	income
1	Denmark	Europe	old	1220
2	Denmark	Europe	young	160
3	France	Europe	old	1450
4	France	Europe	old	1860
5	France	Europe	old	3030
6	France	Europe	old	5880
7	France	Europe	old	6570
8	France	Europe	old	9370
9	France	Europe	young	460
10	France	Europe	young	1300
11	USA	America	old	530
12	USA	America	old	3980
13	USA	America	old	8500
14	USA	America	young	280
15	USA	America	young	700
16	USA	America	young	3640
17	USA	America	young	4680

```
> microdata$ID <- c("Stefan", "Anna", "Nora", "Nora", "David", "Carlos",
                    "Marta", "Katarina", "Stefan", "Helena", "Matteo",
                    "Luca", "Anna", "Erik", "Julia", "Thomas", "Helena")
>
> microdata
    nation continent   age income       ID
1  Denmark    Europe   old   1220   Stefan
2  Denmark    Europe young    160     Anna
3   France    Europe   old   1450     Nora
4   France    Europe   old   1860     Nora
5   France    Europe   old   3030    David
6   France    Europe   old   5880   Carlos
7   France    Europe   old   6570    Marta
8   France    Europe   old   9370 Katarina
9   France    Europe young    460   Stefan
10  France    Europe young   1300   Helena
11     USA   America   old    530   Matteo
12     USA   America   old   3980     Luca
13     USA   America   old   8500     Anna
14     USA   America young    280     Erik
15     USA   America young    700    Julia
16     USA   America young   3640   Thomas
17     USA   America young   4680   Helena
```

Use `contributorVar`

- to specify contributor IDs
- also known as holding indicators

```
> out_4 <- SuppressDominantCells(
      data = microdata,
      formula = ~nation + age * continent,
      dominanceVar =  "income",
      k = c(80, 90),
      contributorVar = "ID",
      allDominance = TRUE
  )
[preAggregate 17*5->16*6]
[extraAggregate 16*6->6*6] Checking .....
GaussSuppression_numttHTT: ....::::
>
> out_4
    nation   age freq income dominant1 dominant2 max1contributor max2contributor n_contr n_non0_contr primary suppressed
1    Total Total   17  53610 0.1747808 0.3363179        Katarina            Anna      13           13   FALSE      FALSE
2  Denmark Total    2   1380 0.8840580 1.0000000          Stefan            Anna       2            2    TRUE       TRUE
3   France Total    8  29920 0.3131684 0.5327540        Katarina           Marta       7            7   FALSE       TRUE
4      USA Total    7  22310 0.3809951 0.5907665            Anna          Helena       7            7   FALSE      FALSE
5    Total   old   10  42390 0.2210427 0.4215617        Katarina            Anna       9            9   FALSE      FALSE
6    Total young    7  11220 0.5329768 0.8573975          Helena          Thomas       6            6   FALSE      FALSE
7  America Total    7  22310 0.3809951 0.5907665            Anna          Helena       7            7   FALSE      FALSE
8   Europe Total   10  31300 0.2993610 0.5092652        Katarina           Marta       8            8   FALSE      FALSE
9  America   old    3  13010 0.6533436 0.9592621            Anna            Luca       3            3    TRUE       TRUE
10  Europe   old    7  29380 0.3189244 0.5425459        Katarina           Marta       6            6   FALSE       TRUE
11 America young    4   9300 0.5032258 0.8946237          Helena          Thomas       4            4   FALSE       TRUE
12  Europe young    3   1920 0.6770833 0.9166667          Helena          Stefan       3            3    TRUE       TRUE
```

allDominance = TRUE

- to see more information in output

Interval suppression

```
lpPackage = "highs"
lpPackage = "Rsymphony"
lpPackage = "Rglpk"
lpPackage = "lpSolve"
```

- Extra suppression with another algorithm
- Time-consuming

```r
SuppressDominantCells(
   data = microdata,
   formula = ~nation + age * continent,
   dominanceVar =  "income",
   pPercent = 30,
   lpPackage = "highs"
)
```

	nation	age	income	lo	up
1	Total	Total	53610	NA	NA
2	Denmark	Total	1380	0	31300
3	France	Total	29920	NA	NA
4	USA	Total	22310	NA	NA
5	Total	old	42390	NA	NA
6	Total	young	11220	NA	NA
7	America	Total	22310	NA	NA
8	Europe	Total	31300	NA	NA
9	America	old	13010	11090	22310
10	Europe	old	29380	NA	NA
11	America	young	9300	0	11220
12	Europe	young	1920	0	11220

```r
SuppressDominantCells(
  data = microdata,
  formula = ~nation + age * continent,
  dominanceVar =  "income",
  pPercent = 30,
  lpPackage = "highs",
  protectionIntervals = TRUE
)
```

	nation	age	income	lo_1	up_1
1	Total	Total	53610	NA	NA
2	Denmark	Total	1380	0	31300
3	France	Total	29920	NA	NA
4	USA	Total	22310	NA	NA
5	Total	old	42390	NA	NA
6	Total	young	11220	NA	NA
7	America	Total	22310	NA	NA
8	Europe	Total	31300	NA	NA
9	America	old	13010	11090	22310
10	Europe	old	29380	NA	NA
11	America	young	9300	0	11220
12	Europe	young	1920	0	11220

```r
SuppressDominantCells(
  data = microdata,
  formula = ~nation + age * continent,
  dominanceVar =  "income",
  pPercent = 30,
  lpPackage = "highs",
  protectionIntervals = TRUE
)
```

	nation	age	income	lomax_income	upmin_income	lo_1	up_1
1	Total	Total	53610	NA	NA	NA	NA
2	Denmark	Total	1380	1014	1746	0	31300
3	France	Total	29920	NA	NA	NA	NA
4	USA	Total	22310	NA	NA	NA	NA
5	Total	old	42390	NA	NA	NA	NA
6	Total	young	11220	NA	NA	NA	NA
7	America	Total	22310	NA	NA	NA	NA
8	Europe	Total	31300	NA	NA	NA	NA
9	America	old	13010	10990	15030	11090	22310
10	Europe	old	29380	NA	NA	NA	NA
11	America	young	9300	8876	9724	0	11220
12	Europe	young	1920	1690	2150	0	11220

```r
SuppressDominantCells(
  data = microdata,
  formula = ~nation + age * continent,
  dominanceVar =  "income",
  pPercent = 30,
  lpPackage = "highs",
  protectionIntervals = TRUE
)
```

	nation	age	income	lomax_income	upmin_income	lo_1	up_1	lo	up	suppressed_integer
1	Total	Total	53610	NA	NA	NA	NA	NA	NA	0
2	Denmark	Total	1380	1014	1746	0	31300	0	53610	1
3	France	Total	29920	NA	NA	NA	NA	NA	NA	2
4	USA	Total	22310	NA	NA	NA	NA	NA	NA	3
5	Total	old	42390	NA	NA	NA	NA	NA	NA	0
6	Total	young	11220	NA	NA	NA	NA	NA	NA	0
7	America	Total	22310	NA	NA	NA	NA	NA	NA	4
8	Europe	Total	31300	NA	NA	NA	NA	NA	NA	3
9	America	old	13010	10990	15030	11090	22310	0	42390	1
10	Europe	old	29380	NA	NA	NA	NA	NA	NA	2
11	America	young	9300	8876	9724	0	11220	0	11220	1
12	Europe	young	1920	1690	2150	0	11220	0	11220	1

```r
SuppressDominantCells(
    data = microdata,
    formula = ~nation + age * continent,
    dominanceVar =  "income",
    pPercent = 30,
    lpPackage = "highs",
    protectionIntervals = TRUE
)
```

	nation	age	income	lomax_income	upmin_income	lo_1	up_1	lo	up	suppressed_integer	primary	suppressed
	Total	Total	53610	NA	NA	NA	NA	NA	NA	0	FALSE	FALSE
	Denmark	Total	1380	1014	1746	0	31300	0	53610	1	TRUE	TRUE
	France	Total	29920	NA	NA	NA	NA	NA	NA	2	FALSE	TRUE
	USA	Total	22310	NA	NA	NA	NA	NA	NA	3	FALSE	TRUE
	Total	old	42390	NA	NA	NA	NA	NA	NA	0	FALSE	FALSE
	Total	young	11220	NA	NA	NA	NA	NA	NA	0	FALSE	FALSE
	America	Total	22310	NA	NA	NA	NA	NA	NA	4	FALSE	TRUE
	Europe	Total	31300	NA	NA	NA	NA	NA	NA	3	TRUE	TRUE
	America	old	13010	NA	NA	NA	NA	NA	NA	1	FALSE	TRUE
	Europe	old	29380	NA	NA	NA	NA	NA	NA	2	TRUE	TRUE
		young	9300	10990	15030	11090	22310	0	42390	1	TRUE	TRUE
			1920	NA	NA	NA	NA	NA	NA	1	TRUE	TRUE
				8876	9724	0	11220	0	11220			
				1690	2150	0	11220	0	11220			

Statistisk sentralbyrå
Statistics Norway

Alternatives to global protection

SuppressLinkedTables()

Gauss algorithm = global protection
against exact disclosure

- Global protection
  - Prevents disclosures that exploit all tables simultaneously
- But Gauss protects only against exact disclosure
  - not interval protection

Consistent protection: alternative to global protection

- Common cells are suppressed equally across tables
- Weaker protection, but not as computationally demanding

- Back-tracking: A common iterative approach

- Consistent protection without iterations in the GaussSuppression package
  - Several variants
  - Methods: `"consistent"` and `"super-consistent"`
  - Parameters: `recordAware` and `collapseAware`

```r
SuppressSmallCounts(
  data = microdata,
  formula = list(tableA = ~nation,
                 tableB = ~age * continent),
  maxN = 3,
  linkedGauss = "consistent"
)
```

	nation	age	freq	primary	suppressed
1	Total	Total	17	FALSE	FALSE
2	Denmark	Total	2	TRUE	TRUE
3	France	Total	8	FALSE	TRUE
4	USA	Total	7	FALSE	FALSE
5	Total	old	10	FALSE	FALSE
6	Total	young	7	FALSE	FALSE
7	America	Total	7	FALSE	FALSE
8	Europe	Total	10	FALSE	FALSE
9	America	old	3	TRUE	TRUE
10	Europe	old	7	FALSE	TRUE
11	America	young	4	FALSE	TRUE
12	Europe	young	3	TRUE	TRUE

```r
SuppressLinkedTables(
  data = microdata,
  fun = SuppressSmallCounts,
  withinArg = list(table_1 = list(dimVar = c("nation")),
                   table_2 = list(dimVar = c("age", "continent"))),
  maxN = 3,
  linkedGauss = "super-consistent"
)
```

```
$table_1
   nation freq primary suppressed
1   Total   17   FALSE      FALSE
2 Denmark    2    TRUE       TRUE
3  France    8   FALSE      FALSE
4     USA    7   FALSE       TRUE
```

```
$table_2
    age continent freq primary suppressed
1 Total     Total   17   FALSE      FALSE
2 Total   America    7   FALSE       TRUE
3 Total    Europe   10   FALSE       TRUE
4   old     Total   10   FALSE      FALSE
5   old   America    3    TRUE       TRUE
6   old    Europe    7   FALSE       TRUE
7 young     Total    7   FALSE      FALSE
8 young   America    4   FALSE       TRUE
9 young    Europe    3    TRUE       TRUE
```

```
> out_5 <- SuppressLinkedTables(
    data = microdata,
    fun = SuppressDominantCells,
    withinArg = list(table_1 = list(dimVar = c("nation"), k = c(80, 90)),
                     table_2 = list(dimVar = c("age", "continent"), pPercent = 150)),
    dominanceVar =  "income",
    contributorVar = "ID",
    lpPackage = "highs",
    protectionIntervals = TRUE
  )
[preAggregate 17*12->16*11]
[extraAggregate 16*11->3*11] Checking .....
[preAggregate 17*12->16*12]
[extraAggregate 16*12->4*12] Checking .....

====== Linked GaussSuppression by "super-consistent" algorithm:

GaussSuppression_numttHTT: ........
(7*2-0exact->6*1-GaussI->6*1)

Using highs for intervals...
------
(7*2-DDrow->6*2->)
..
2+1-
  1: 1 new, (29920.0) 1+
GaussSuppression_none: ....
(7*1-0exact->7*1-GaussI->7*1)

Using highs for intervals...
------
```

- Interval protection and consistent protection algorithms can be combined

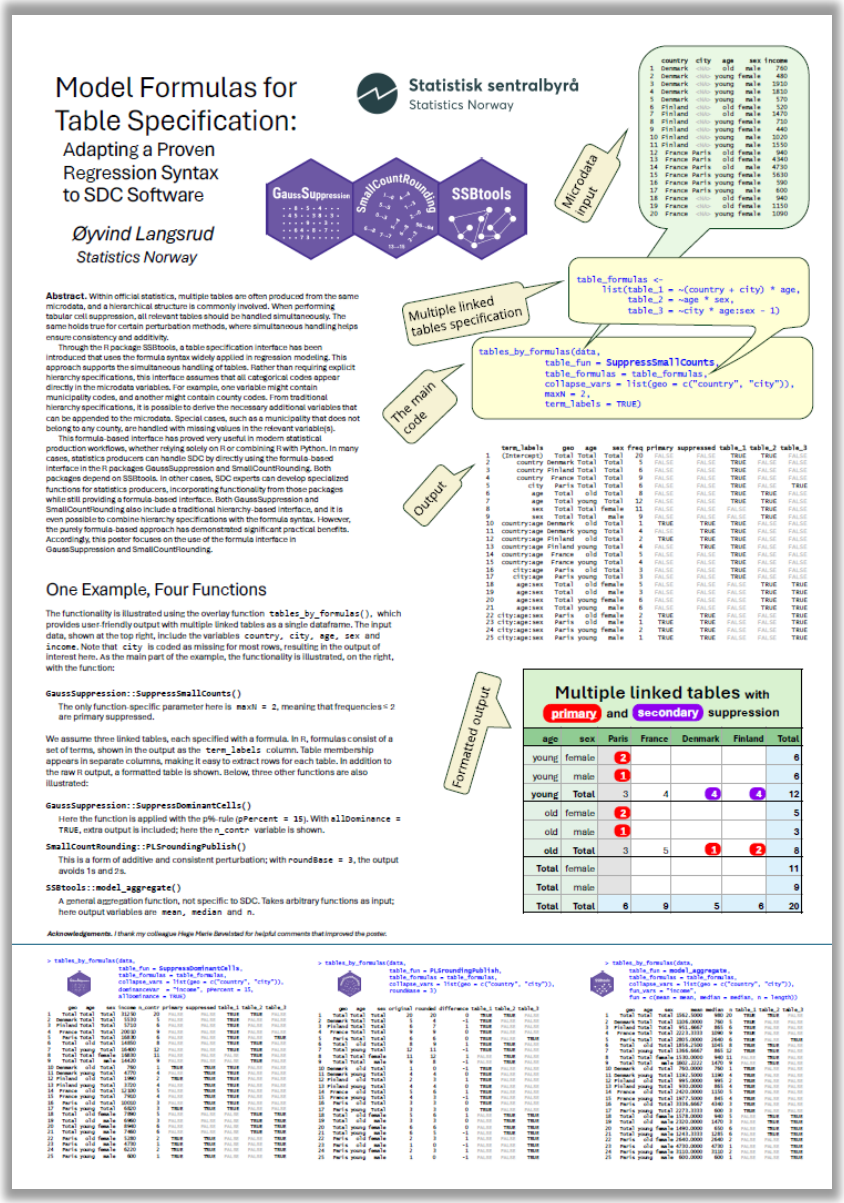
```
> out_5
$table_1
   nation freq income lomax_income upmin_income lo_1  up_1 lo    up suppressed_integer primary suppressed
1   Total   17  53610           NA           NA   NA    NA NA    NA                  0   FALSE      FALSE
2 Denmark    2   1380     1226.667     1533.333    0 23690  0 53610                  1    TRUE       TRUE
3  France    8  29920           NA           NA   NA    NA NA    NA                  3   FALSE       TRUE
4     USA    7  22310           NA           NA   NA    NA NA    NA                  2   FALSE       TRUE

$table_2
    age continent freq income lomax_income upmin_income lo_1  up_1 lo    up suppressed_integer primary sup
1 Total     Total   17  53610           NA           NA   NA    NA NA    NA                  0   FALSE
2 Total   America    7  22310        18690        25930   NA    NA NA    NA                  1    TRUE
3 Total    Europe   10  31300           NA           NA   NA    NA NA    NA                  2   FALSE
4   old     Total   10  42390           NA           NA   NA    NA NA    NA                  2   FALSE
5   old   America    3  13010          790        25230    0 23690  0 53610                  1    TRUE
6   old    Europe    7  29380        28765        29995    0 53610  0 53610                  1    TRUE
7 young     Total    7  11220         3850        18590    0 53610  0 53610                  1    TRUE
8 young   America    4   9300         3260        15340    0 23690  0 53610                  1    TRUE
9 young    Europe    3   1920          130         3710    0 53610  0 53610                  1    TRUE
```

TALL

SOM FORTELLER

tables_by_formulas()



Model Formulas for Table Specification:

Adapting a Proven Regression Syntax to SDC Software

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Microdata input

Multiple linked tables specification

The main code

Output

Formatted output

Multiple linked tables with primary and secondary suppression

One Example, Four Functions

```r
SuppressSmallCounts(
  data = microdata,
  formula = list(tableA = ~nation,
                 tableB = ~age * continent),
  maxN = 3,
  linkedGauss = "consistent"
)
```

	nation	age	freq	primary	suppressed
1	Total	Total	17	FALSE	FALSE
2	Denmark	Total	2	TRUE	TRUE
3	France	Total	8	FALSE	TRUE
4	USA	Total	7	FALSE	FALSE
5	Total	old	10	FALSE	FALSE
6	Total	young	7	FALSE	FALSE
7	America	Total	7	FALSE	FALSE
8	Europe	Total	10	FALSE	FALSE
9	America	old	3	TRUE	TRUE
10	Europe	old	7	FALSE	TRUE
11	America	young	4	FALSE	TRUE
12	Europe	young	3	TRUE	TRUE

```r
tables_by_formulas(
  data = microdata,
  table_fun = SuppressSmallCounts,
  table_formulas = list(tableA = ~nation,
                        tableB = ~age * continent),
  maxN = 3,
  collapse_vars = list(geo = c("nation", "continent"))
)
```

	geo	age	freq	primary	suppressed	tableA	tableB
1	Total	Total	17	FALSE	FALSE	TRUE	TRUE
2	Denmark	Total	2	TRUE	TRUE	TRUE	FALSE
3	France	Total	8	FALSE	TRUE	TRUE	FALSE
4	USA	Total	7	FALSE	FALSE	TRUE	FALSE
5	Total	old	10	FALSE	FALSE	FALSE	TRUE
6	Total	young	7	FALSE	FALSE	FALSE	TRUE
7	America	Total	7	FALSE	FALSE	FALSE	TRUE
8	Europe	Total	10	FALSE	FALSE	FALSE	TRUE
9	America	old	3	TRUE	TRUE	FALSE	TRUE
10	Europe	old	7	FALSE	TRUE	FALSE	TRUE
11	America	young	4	FALSE	TRUE	FALSE	TRUE
12	Europe	young	3	TRUE	TRUE	FALSE	TRUE

```r
tables_by_formulas(
  data = microdata,
  table_fun = SSBtools::model_aggregate,
  table_formulas = list(tableA = ~nation,
                        tableB = ~age * continent),
  collapse_vars = list(geo = c("nation", "continent")),
  fun_vars = "income",
  fun = c(mean = mean, median = median, n = length)
)
```

	geo	age	mean	median	n	tableA	tableB
1	Total	Total	3153.529	1860	17	TRUE	TRUE
2	Denmark	Total	690.000	690	2	TRUE	FALSE
3	France	Total	3740.000	2445	8	TRUE	FALSE
4	USA	Total	3187.143	3640	7	TRUE	FALSE
5	Total	old	4239.000	3505	10	FALSE	TRUE
6	Total	young	1602.857	700	7	FALSE	TRUE
7	America	Total	3187.143	3640	7	FALSE	TRUE
8	Europe	Total	3130.000	1655	10	FALSE	TRUE
9	America	old	4336.667	3980	3	FALSE	TRUE
10	Europe	old	4197.143	3030	7	FALSE	TRUE
11	America	young	2325.000	2170	4	FALSE	TRUE
12	Europe	young	640.000	460	3	FALSE	TRUE

```r
tables_by_formulas(
  data = microdata,
  table_fun = SuppressSmallCounts,
  table_formulas = list(tableA = ~nation,
                        tableB = ~age * continent),
  maxN = 3,
  collapse_vars = list(geo = c("nation", "continent")),
)
```

	geo	age	freq	primary	suppressed	tableA	tableB
1	Total	Total	17	FALSE	FALSE	TRUE	TRUE
2	Denmark	Total	2	TRUE	TRUE	TRUE	FALSE
3	France	Total	8	FALSE	TRUE	TRUE	FALSE
4	USA	Total	7	FALSE	FALSE	TRUE	FALSE
5	Total	old	10	FALSE	FALSE	FALSE	TRUE
6	Total	young	7	FALSE	FALSE	FALSE	TRUE
7	America	Total	7	FALSE	FALSE	FALSE	TRUE
8	Europe	Total	10	FALSE	FALSE	FALSE	TRUE
9	America	old	3	TRUE	TRUE	FALSE	TRUE
10	Europe	old	7	FALSE	TRUE	FALSE	TRUE
11	America	young	4	FALSE	TRUE	FALSE	TRUE
12	Europe	young	3	TRUE	TRUE	FALSE	TRUE

```r
tables_by_formulas(
  data = microdata,
  table_fun = SuppressSmallCounts,
  table_formulas = list(tableA = ~nation,
                        tableB = ~age * continent),
  maxN = 3,
  collapse_vars = list(geo = c("nation", "continent")),
  da_vars = "income", da_fun = c(mean = mean, median = median)
)
```

	geo	age	freq	mean	median	primary	suppressed	tableA	tableB
1	Total	Total	17	3153.529	1860	FALSE	FALSE	TRUE	TRUE
2	Denmark	Total	2	690.000	690	TRUE	TRUE	TRUE	FALSE
3	France	Total	8	3740.000	2445	FALSE	TRUE	TRUE	FALSE
4	USA	Total	7	3187.143	3640	FALSE	FALSE	TRUE	FALSE
5	Total	old	10	4239.000	3505	FALSE	FALSE	FALSE	TRUE
6	Total	young	7	1602.857	700	FALSE	FALSE	FALSE	TRUE
7	America	Total	7	3187.143	3640	FALSE	FALSE	FALSE	TRUE
8	Europe	Total	10	3130.000	1655	FALSE	FALSE	FALSE	TRUE
9	America	old	3	4336.667	3980	TRUE	TRUE	FALSE	TRUE
10	Europe	old	7	4197.143	3030	FALSE	TRUE	FALSE	TRUE
11	America	young	4	2325.000	2170	FALSE	TRUE	FALSE	TRUE
12	Europe	young	3	640.000	460	TRUE	TRUE	FALSE	TRUE

```r
tables_by_formulas(
  data = microdata,
  table_fun = SuppressSmallC
  table_formulas = list(tableA =
                        tableB =
  maxN = 3,
  collapse_vars = list(geo = c("nation", "continent")),
  da_vars = "income", da_fun = c(mean = mean, median = median)
)
```

	geo	age	freq	mean	median	primary	suppressed	tableA	tableB
1	Total	Total	17	3153.529	1860	FALSE	FALSE	TRUE	TRUE
2	Denmark	Total	2	690.000	690	TRUE	TRUE	TRUE	FALSE
3	France	Total	8	3740.000	2445	FALSE	TRUE	TRUE	FALSE
4	USA	Total	7	3187.143	3640	FALSE	FALSE	TRUE	FALSE
5	Total	old	10	4239.000	3505	FALSE	FALSE	FALSE	TRUE
6	Total	young	7	1602.857	700	FALSE	FALSE	FALSE	TRUE
7	America	Total	7	3187.143	3640	FALSE	FALSE	FALSE	TRUE
8	Europe	Total	10	3130.000	1655	FALSE	FALSE	FALSE	TRUE
9	America	old	3	4336.667	3980	TRUE	TRUE	FALSE	TRUE
10	Europe	old	7	4197.143	3030	FALSE	TRUE	FALSE	TRUE
11	America	young	4	2325.000	2170	FALSE	TRUE	FALSE	TRUE
12	Europe	young	3	640.000	460	TRUE	TRUE	FALSE	TRUE

Functionality that corresponds to SSBtools::model_aggregate()

- News for GaussSuppression 1.2.3
- ?SuppressDominantCells
  - The bottom examples

Final remarks

- *GaussSuppression* is a comprehensive R package
- Many features are shown
- Much more exists
  - The singleton problem in magnitude tables
  - The problem of zeros in frequency tables
  - K-disclosure suppression
  - ...
  -

Some references and links

- CRAN: *Vignettes and News*
- The R Journal 2023: *Sparse Model Matrices for Multidimensional Hierarchical Aggregation*
- PSD 2024: *Secondary Cell Suppression by Gaussian Elimination: An Algorithm Suitable for Handling Issues with Zeros and Singletons*
- Expert Meeting 2025: *Model Formulas for Table Specification: Adapting a Proven Regression Syntax to SDC Software*
- INSEE JMS 2025: *Tabular data protection with the R-package GaussSuppression: from algorithm to application*